

Event Synthesis for Light Field Videos using Recurrent Neural Networks

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ABSTRACT

Light field videos (LFVs), consisting of multiple angles of view, yield higher complexity in performing computer vision tasks. The emerging event cameras offer a new means for light-weight processing of LFVs, but it is infeasible to build an LFV capturing device with multiple event cameras due to their high costs. In this poster, we propose a novel “event synthesis for light field videos” (ES4LFV) model by using recurrent neural networks and build a preliminary dataset for training. The ES4LFV can synthesize events for light field video (E-LFV) from LFV camera array and single event camera. The experimental results show that ES4LFV outperforms the traditional method by 3.1dB in PSNR.

Index Terms: Light Field Image—Event-based Vision—RNN;

1 INTRODUCTION

The 6 degree-of-freedom (6 DoF) experience has become the developing trend for future VR video with enhanced immersion. Light field videos (LFVs), consisting of sub-views captured from multiple angles, naturally enable 6 DoF VR video experience with their extra angular information. However, LFVs also impose significant challenges in performing computer vision tasks, e.g., object tracking, due to their dense multi-view structure and bulky video sizes [4]. Moreover, the framerate of LFVs is typically 30 fps, which is insufficient for some demanding tasks (e.g., tracking fast-moving objects like bullets). Thanks to the emerging “event” sensing technology, above challenges can address in a new way. Unlike traditional cameras, an event camera only captures the local pixel-wise brightness changes independently and asynchronously. An “event” is defined as a tuple (t, x, y, p) where (x, y) determines the coordinate of pixel, t is the timestamp and $p \in \{-1, +1\}$ is the event polarity. Event cameras support up to 1 million fps temporal resolution (compared to 30 fps for regular LFV cameras [1]), up to 120 dB dynamic range (compared to 40-60 dB for regular cameras), typically around 1Mb/s data rate and very low power consumption (around 1mW) [3]. As a result, event processing enables low-power computer vision tasks in the multiple sub-views of LFVs at an extremely high temporal resolution.

To realize event processing on LFVs, however, there are a couple of problems to be resolved. Firstly, an event camera costs several thousand USDs, it is impractical to build an camera array consisting of tens of event cameras. Secondly, for event signals synchronized for LFVs, there is no relevant dataset publicly available and hence the well-studied deep-learning-based algorithms cannot be directly applied to event processing for LFVs. Therefore, it is highly desirable to develop an algorithm with the capability

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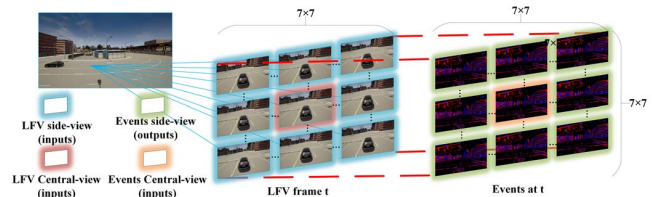


Figure 1: Virtual 7×7 LFV camera array and 7×7 event camera array are synchronized for capturing frames and events simultaneously.

of synthesizing multi-view event signals from a LFV camera array and just one event camera. In the literature, the central view augmentation propagation (CVAP) method [6, 5] could propagate the information from one single sub-view to all the remaining sub-views, which may be adapted to event synthesis for LFVs, but this method neglects the temporal information in LFVs and therefore may not result in desirable results.

To resolve the above problems, in this poster we propose a novel “event synthesis for light field video” (ES4LFV) algorithm based on recurrent neural network (RNN) and compared its performance with a baseline method adapted from the CVAP (see Section 3.1 for details). Due to the infeasibility in constructing an event camera array in reality, we build a virtual event camera array synchronized with LFV camera array in the Carla Simulator virtual environment. We then constructed a preliminary dataset of “event for LFVs” (E-LFVs), train our model and conduct experiments on this dataset. We expect that the trained model can be adapted to real-world LFVs too. The experimental results show that proposed ES4LFV outperforms the CVAP-adapted synthesis by 3.1dB in PSNR and also yields more pleasant visual experience. We believe that our work is the first one focusing on E-LFV with dedicatedly constructed E-LFV dataset and RNN-based synthesis model.

2 METHODOLOGY

To realize ES4LFV and train a sophisticated model, we need to firstly construct a dataset containing both LFVs and event signals for training. As mentioned in Section 1, due to the difficulty in building an event camera array in reality, we adopt the Carla virtual environment and setup synchronized 7×7 frame camera array and 7×7 event camera array in Carla as Figure 1. By this means, we will have 7×7 LFVs with corresponding events as ground truth. At this preliminary stage, we have captured 50 pairs of LFVs (30 fps) and corresponding event signals (90 fps) in Carla simulator. In the E-LFV synthesis process, only the events of the central-view is used to predict the events of all the remaining sub-views. The captured events for other side-views, on the other hand, are only used as ground-truth for our model training and testing.

The flowchart of our proposed ES4LFV model is shown in Figure 2. In our experiment setup, the LFV is 30 fps and events are 90 fps, and 6 event frames are used for predicting. As a result, the input data contains 2 frames of LFV (i.e., 2 LFIs) and 6 frames of events. During each epoch, we only choose a

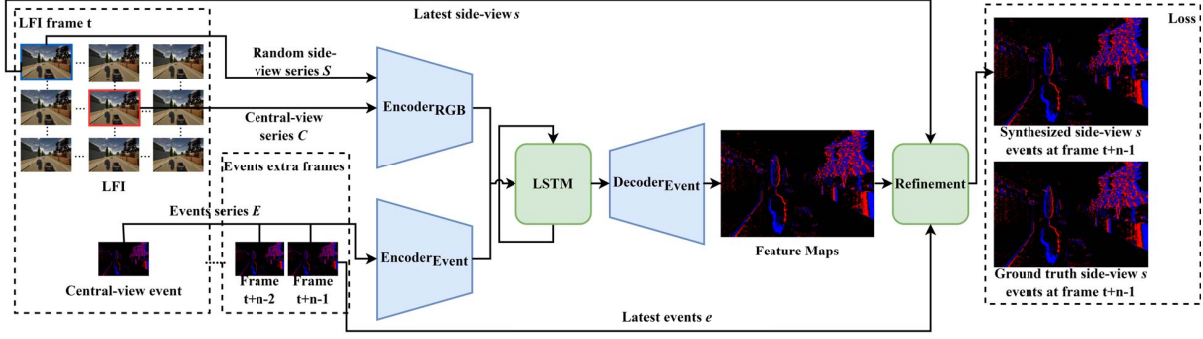


Figure 2: Our proposed ES4LFV model. Events have “extra frames” because event camera has higher framerate than LFV camera. n stands for the framerate ratio, hence $t+n-1$ is the last frame before there is a corresponding LFV frame. For example, in our experiments, LFV is 30 fps and events are 90 fps, then $n = 3$. If we choose window length of 6 timeframes, then we have 2 LFV frames and 6 event frames as inputs.

Table 1: PSNR comparison for the proposed ES4LFV and the CVAP-adapted synthesis.

PSNR	CVAP-adapted	ES4LFV
video1	32.77	33.24
video2	32.95	37.45
...	...	...
All testing pairs (10 in total)	32.46	35.58

random side-view s to leverage the training load. Information from three series of data - LFV central-view c , LFV side-view s and central-view events - will be firstly extracted by image encoder [2] into $2 \times 376 \times 541 \times 16$ and $6 \times 376 \times 541 \times 16$ features (first dimension is the number of frames). Then LSTM will further extract temporal information and output $376 \times 541 \times 32$ features. After concatenation and decoding [2], the outputs become the $376 \times 541 \times 3$ intermediate feature map containing both event signal information and LFV parallax information. At this stage, we obtain feature maps, another refinement network is used to produce final outputs. Inputs of refinement network are $376 \times 541 \times 3$ feature map, latest side-aperture s and latest events e . Two 3×3 Conv2D layers and a Dense layer produce the desired visualized events. The model loss is calculated by the mean squared error between our synthesized events and real events from simulator.

3 EXPERIMENTAL RESULTS AND ANALYSIS

3.1 CVAP-Adapted LFV Synthesis

CVAP-adapted method is proposed in [6, 5]. We input LFV frame (i.e., LFI) and central-view events as image and output the event signals for LFI. However, CVAP-adapted method has its limitation when events do not have corresponding LFV frame. In this case, the nearest LFV frame is used for predicting.

3.2 Experimental Results

In our experiments, we use 40 pairs of LFVs and events for training and remaining 10 pairs for testing, and compare the PSNR values (average of all frames and all sub-views) between our synthesized events and the ground-truth events. The experimental results are listed in Table 1. We can observe that our proposed ES4LFV significantly outperforms the CVAP-adapted synthesis by 3.1dB in PSNR. Some sample of the synthesized events are shown in Figure 3, where Figure 3(d)-(f) show that the CVAP-adapted synthesis has produced some obvious artifacts while the output from the proposed ES4LFV are visually more pleasant. We think the reasons of the artifacts from CVAP-adapted method is due to texture information of LFV frames being wrongly included when there lacks of significant brightness change within certain part of the image.

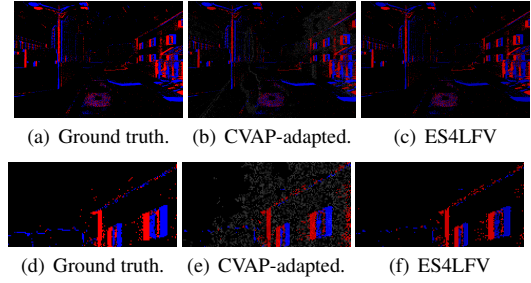


Figure 3: Samples of synchronized events (left-top corner of the last event frame). (d)-(f) are enlarged view of (a)-(c). We can observe wrong parallax estimation

4 CONCLUSION

We propose a novel ES4LFV model that can synthesize E-LFV from a LFV camera array and single event camera. With ES4LFV, some computer vision tasks such as fast-object tracking in LFVs can be performed in the event domain with low-complexity and high temporal resolution. However, to overcome the limitations of the current ES4LFV, our future works would include: 1) further investigating the correlations between temporal and angular information of LFVs; 2) further improving our model by using more advanced RNN model to handle blurry scenes and speedy objects; 3) significantly enlarging our dataset for better training and testing.

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